Gene Therapy

MBGT309

Prof. Dr. Tamer Roshdy

Professor of Molecular Cell Biology




صباح
السعادة
والهنا

ميثـــاق الجلســـات



الابتسامة مفتاح السعادة



الوقت كالسيف



العمل الجماعي اساس النجاح



كن قدوة لغيرك



من فضلك اغلق الهاتف



الحديث الجانبي يشتت الانتباه



الاختلاف فى الرأي لا يفسد للود قضية



focus

What is Gene Therapy?

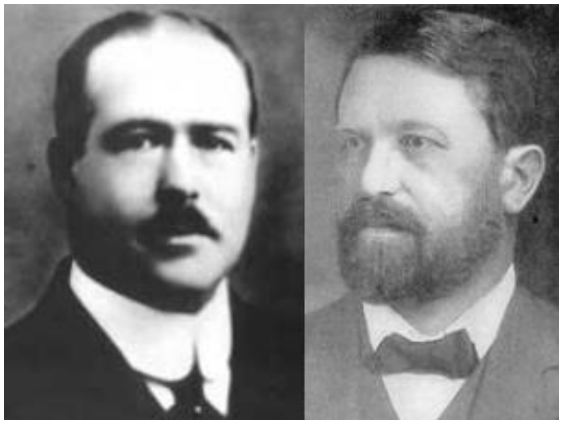
- An approach of treating diseases by either modifying the expressions of an individual's genes or correction of abnormal genes.

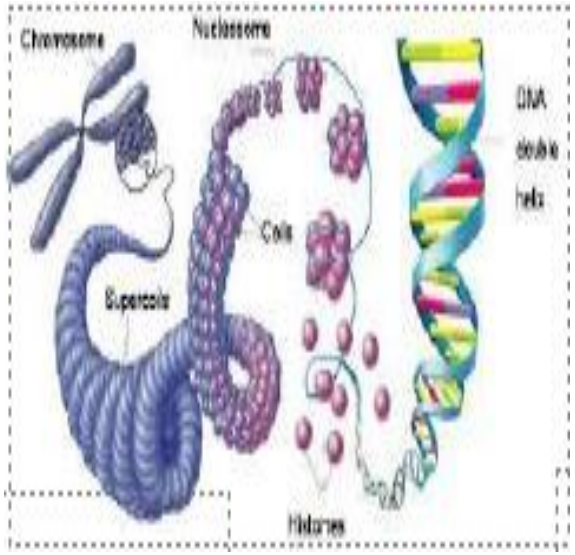
- Genes as Drugs: It was the logical consequence of at least two major advancements that were occurring in those years, namely the exponentially growing knowledge of human gene function and the impact of their mutations, and the development of progressively more effective technologies for the delivery of DNA into mammalian cells.

History



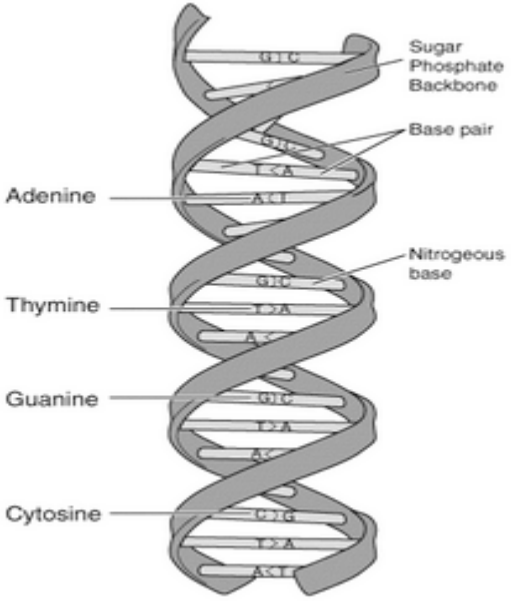
- **Mendelian inheritance** (also known as **Mendelism**) is a type of biological inheritance following the principles originally proposed by Gregor Mendel in 1865 and 1866, re-discovered in 1900 by Hugo de Vries and Carl Correns, and later popularized by William Bateson.[1] These principles were initially controversial. When Mendel's theories were integrated with the Boveri–Sutton chromosome theory of inheritance by Thomas Hunt Morgan in 1915, they became the core of classical genetics. Ronald Fisher combined these ideas with the theory of natural selection in his 1930 book *The Genetical Theory of Natural Selection*, putting evolution onto a mathematical footing and forming the basis for population genetics within the modern evolutionary synthesis





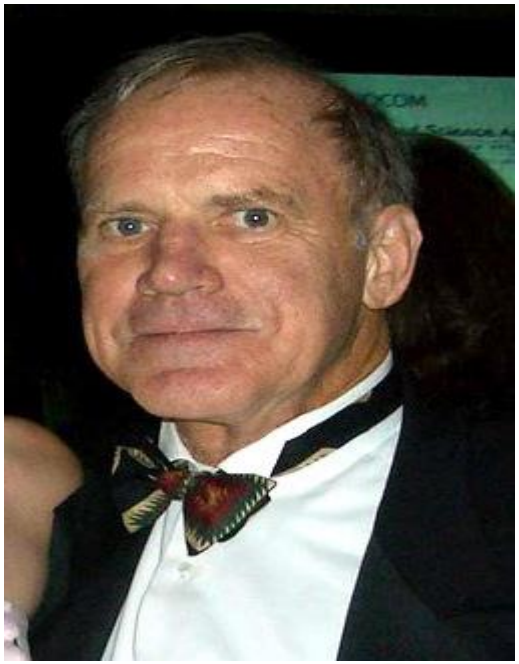
- The chromosome theory of inheritance is credited to papers by Walter Sutton in 1902[and 1903,[as well as to independent work by Theodor Boveri during roughly the same periodThe **Boveri–Sutton chromosome theory** (also known as the **chromosome theory of inheritance** or the **Sutton–Boveri theory**) is a fundamental unifying theory of genetics which identifies chromosomes as the carriers of genetic material.[1][2][3] It correctly explains the mechanism underlying the laws of Mendelian inheritance by identifying chromosomes with the paired factors (particles) required by Mendel's laws. It also states that chromosomes are linear structures with genes located at specific sites called loci along them

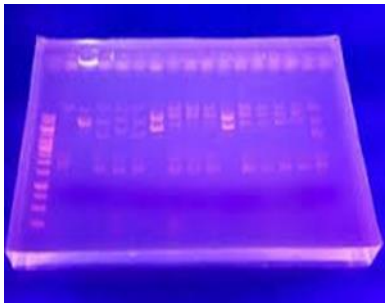
- **"Molecular Structure of Nucleic Acids: A Structure for Deoxyribose Nucleic Acid"** was the first article published to describe the discovery of the double helix structure of DNA, using X-ray diffraction and the mathematics of a helix transform. It was published by Francis Crick and James D. Watson in the scientific journal *Nature* on pages 737–738 of its 171st volume (dated 25 April 1953)





- in 1952 plasmids discovered by **Lederberg** as a transferring the genetic information between different cells
- In 1967 ligase discovered by **Weiss and**
- **Richardson**
- In 1970 restriction enzymes discovered by **Hamilton smith**
- In 1977 the pioneering development of DNA **sequencing by Frederick Sanger**







- The **polymerase chain reaction** (**PCR**) is a method widely used to make millions to billions of copies of a specific DNA sample rapidly, allowing scientists to amplify a very small sample of DNA (or a part of it) sufficiently to enable detailed study. PCR was invented in 1983 by American biochemist Kary Mullis

History

- **1960's** : The concepts of Gene Therapy was introduced.
- **1972 :** Friedman and Roblin authored a paper in Science titled "Gene therapy for human genetic disease."



- **1984:** A retrovirus vector system was designed that could efficiently insert foreign genes into mammalian chromosomes.

- **1990: The first approved gene therapy in the US took place on 14 September 1990, at the National Institutes of Health (NIH), under the direction of William French Anderson.**



- Four-year-old **Ashanti De Silva** received treatment for a genetic defect that left her with ADA-SCID, a severe immune system deficiency.



- **1992:** Doctor **Claudio Bordignon** performed the first procedure of gene therapy using hematopoietic stem cells as vectors to deliver genes intended to correct hereditary diseases.



- **1999:** Death of **Jesse Gelsinger** in a gene therapy experiment



- **2003 :** a research team inserted genes into the brain for the first time. They used liposomes which, unlike viral vectors, are small enough to cross the blood–brain barrier.
- **2006 :** successful use of gene therapy to treat two adult patients for X-linked chronic granulomatous disease.
- **2007:** first gene therapy trial for inherited retinal disease

- **2010 :** an 18 year old male patient in France with beta-thalassemia major had been successfully treated.
- **2011:** Medical community accepted that it can cure HIV as in 2008, **Gero Hutter** has cured a man from HIV using gene therapy.
- **20011- 2015:** Research is still ongoing and the number of diseases that has been treated successfully by gene therapy increases.

Genes as Drugs

- Genes as Drugs: It was the logical consequence of at least two major advancements that were occurring in those years, namely the exponentially growing knowledge of human gene function and the impact of their mutations, and the development of progressively more effective technologies for the delivery of DNA into mammalian cells.

- The first foreseen applications of gene therapy were for the treatment of patients with inherited monogenic disorders with recessive inheritance. Examples of these disorders are cystic fibrosis, muscular dystrophies, lysosomal storage disorders, hemophilia, and several thousand other different conditions. These diseases are due to single gene defects and the pathologic phenotype becomes evident only when both alleles are defective.

- Most importantly, the relatives of patients carrying one normal copy of the gene that is mutated in the patients are perfectly healthy; thus, should we be able to transfer one normal copy of the disease gene in the affected individuals, we would have found a cure for the disease.

- The development of any gene therapy approach requires an accurate evaluation of a series of parameters, including assessment of the likelihood that a given disease might realistically be treated by gene therapy, followed by the choice of the correct therapeutic gene, regulatory signals, should this gene be expressed in the cells, delivery system, and administration route.

Gene Therapy: An Overview

The development of any gene therapy approach requires an accurate evaluation of a series of parameters, including assessment of the likelihood that a **given disease might realistically be treated by gene therapy**, followed by the **choice of the correct therapeutic gene**, regulatory signals, should this gene be expressed in the cells, **delivery system**, and **administration route.**